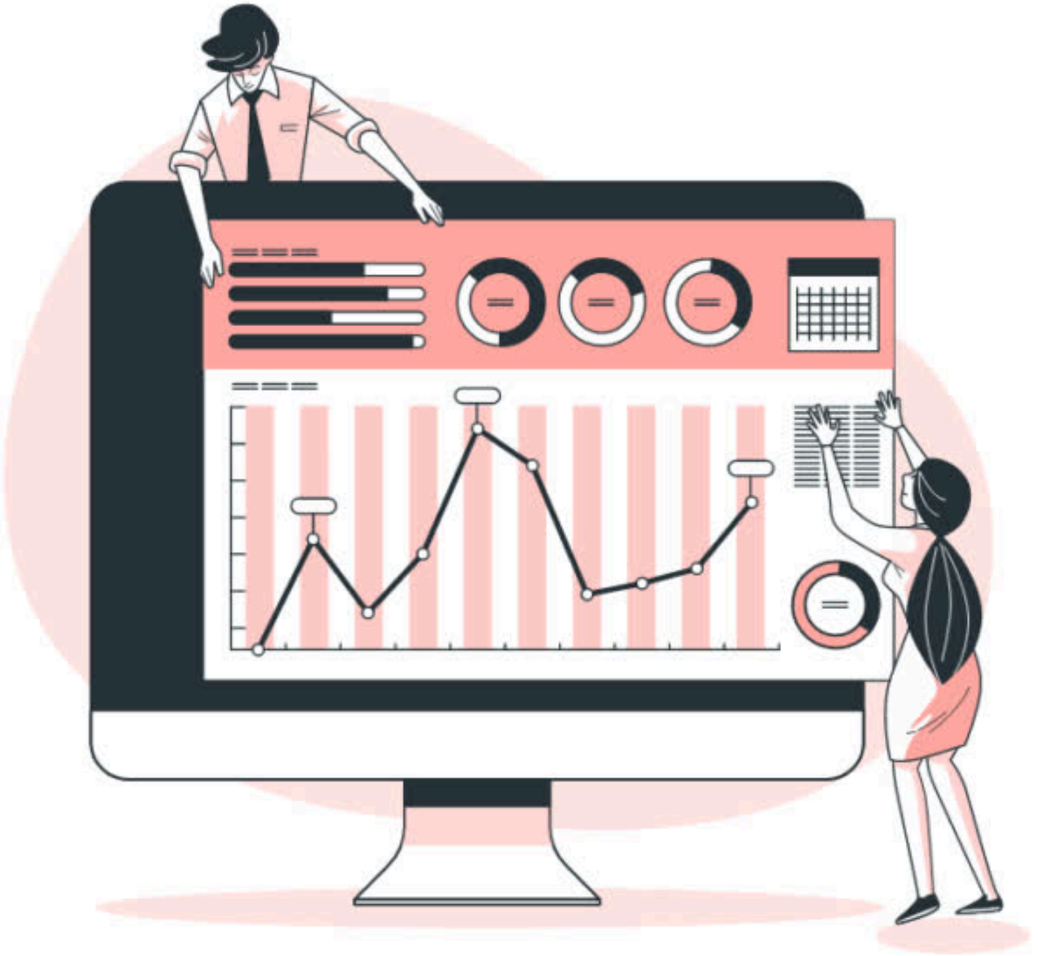
Non-Linear Optimization

Team 10
Aviral Narang, Dongwook Lee,
Flavie Favreaux, Katie Liang



Purpose

Analysis of individual stock portfolios on two political figures using non-linear optimization to maximum reward at a minimal risk.



01

Nancy Pelosi

- Representative from California
- 52nd Speaker of the House of Representative

Stock Portfolio



Stock Portfolio

Shape	
Rows	252
Columns	90

Timeframe

January 2, 2024



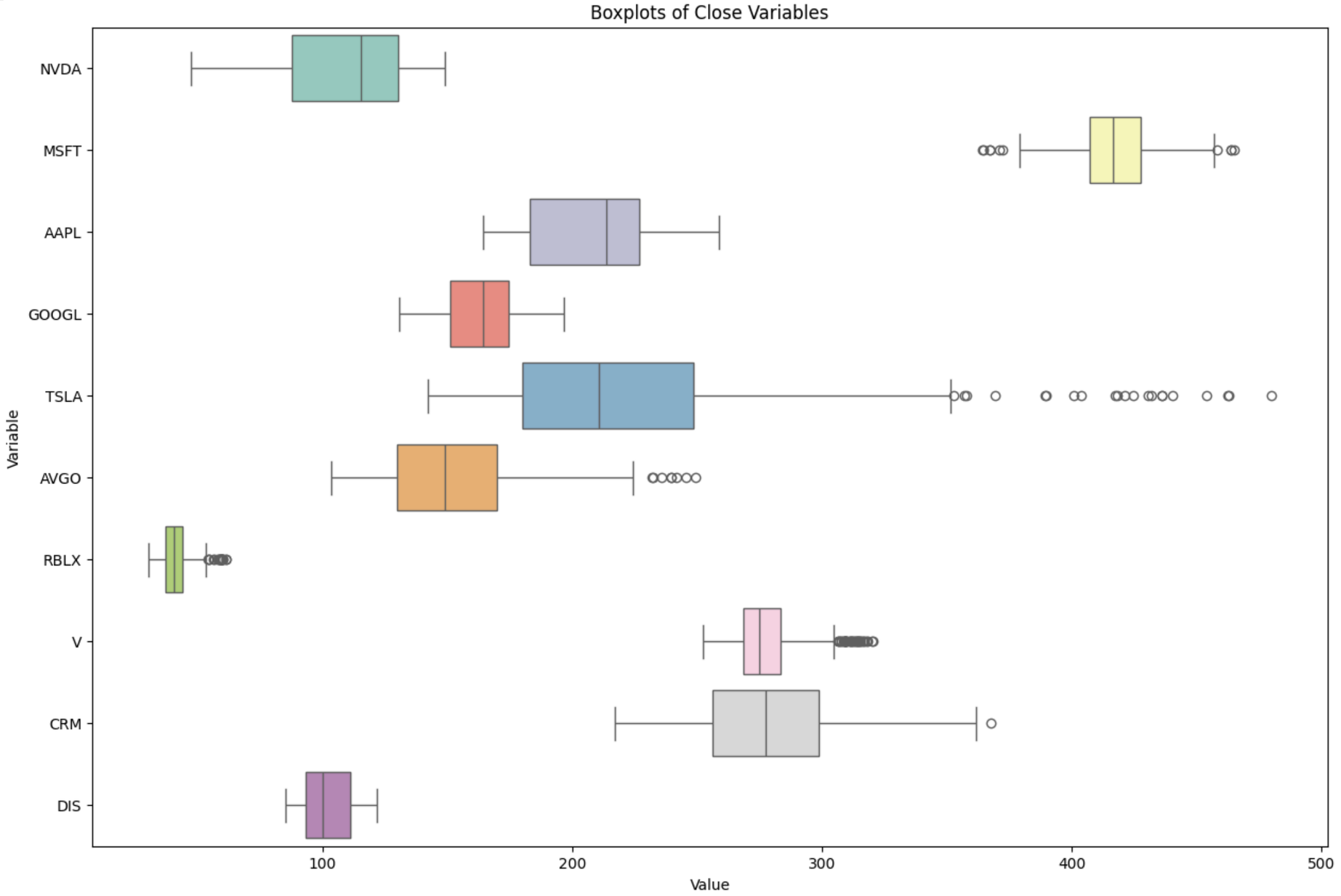
December 31, 2024

Volatility and Price Metrics (Based on Adj Close)

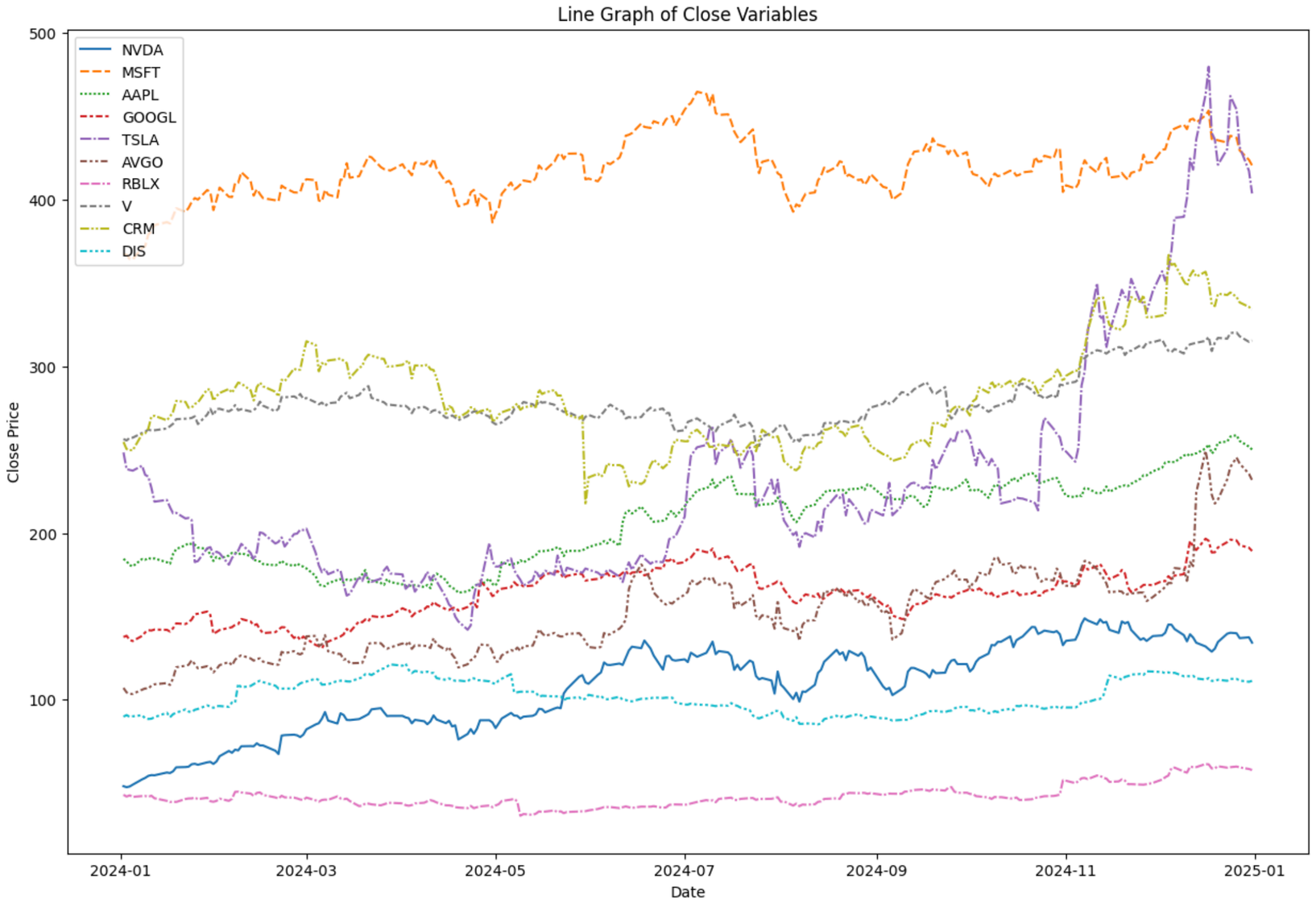
Volatility		
Highest	TSLA	$70.99
Lowest	RBLX	$6.79

Price		
Highest	TSLA	$417.91
Lowest	RBLX	$30.42

Boxplots


Boxplots of Close Variables
NVDA
MSFT
AAPL
GOOGL
TSLA
AVGO
RBLX
V
CRM
DIS
Variable
100
200
300
400
500
Value

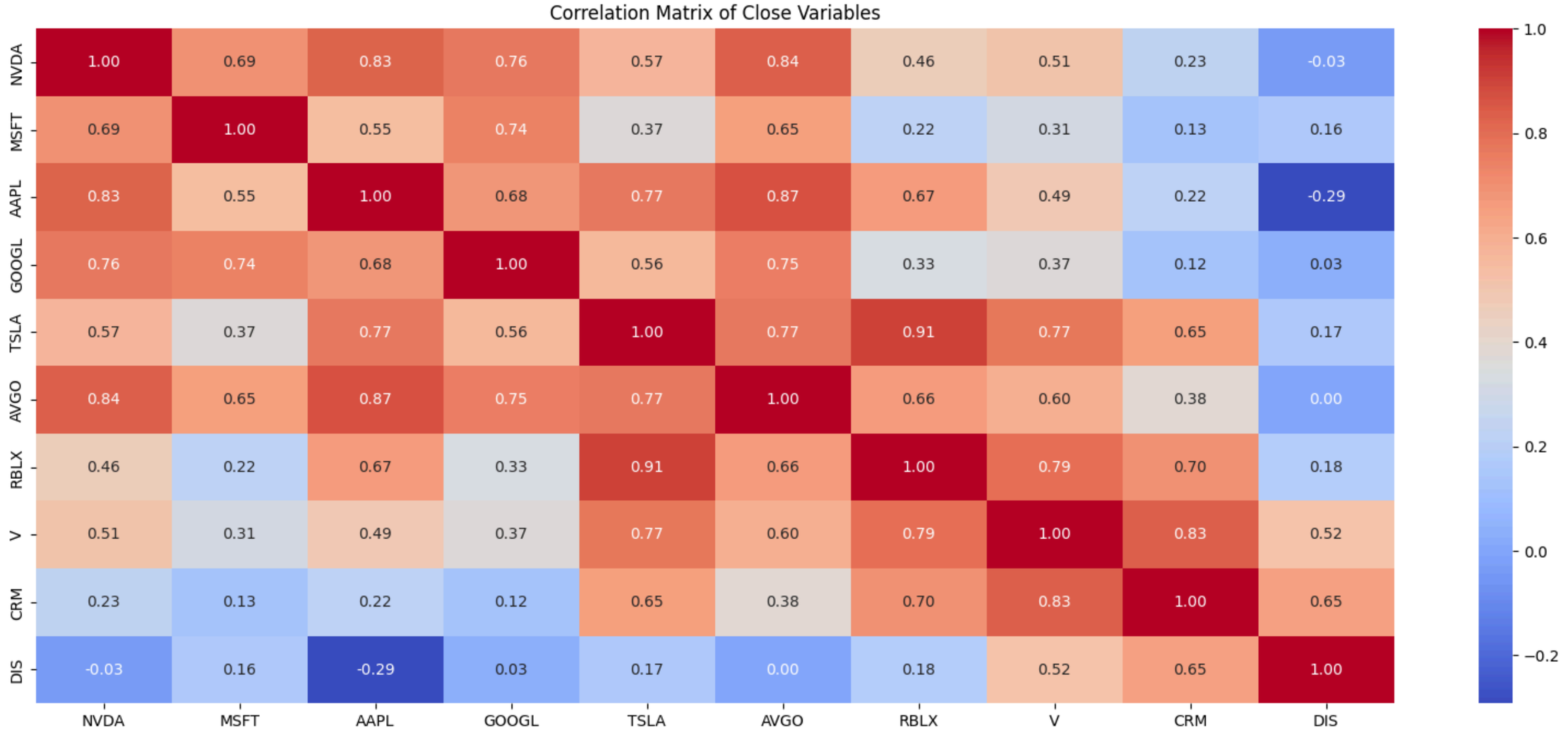
Time Series


Line Graph of Close Variables
NVDA
MSFT
AAPL
GOOGL
TSLA
AVGO
RBLX
V
CRM
DIS
500
400
300
200
100
Close Price
2024-01
2024-03
2024-05
2024-07
2024-09
2024-11
2025-01
Date

Backtracking Portfolio Investment Size



Correlation Matrix



Objective Formula

MAX

108 x NVDA +

418 x MSFT +

206 x AAPL +

163 x GOOGL +

230 x TSLA +

151 x AVGO +

42 x RBLX +

279 x V +

383 x CRM +

101 x DIS

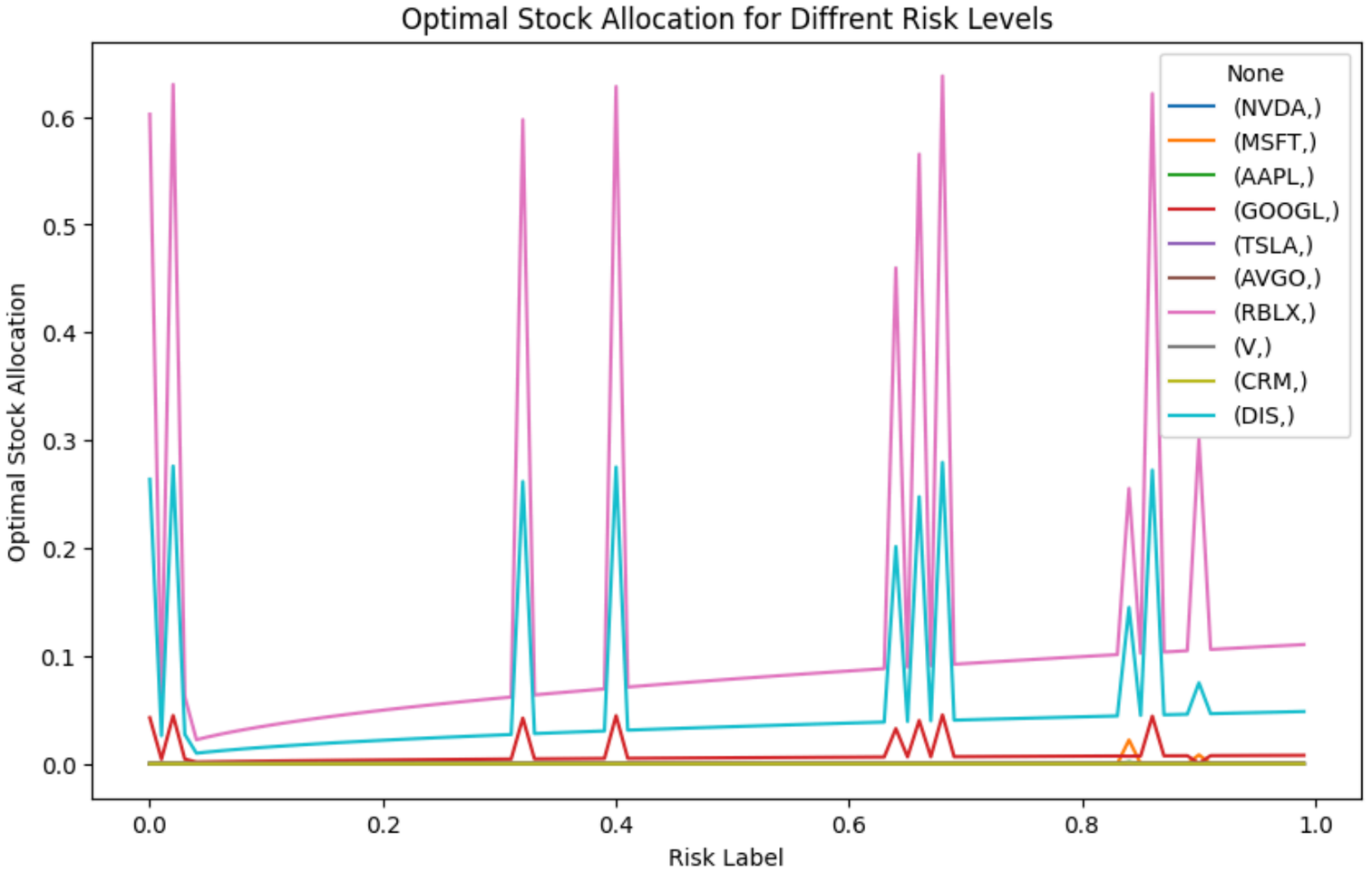
Constraints

m.total_risk = Constraint(expr = m.NVDA + m.MSFT + m.AAPL + m.GOOGL + m.TSLA + m.AVGO + m.RBLX + m.V + m.CRM m.DIS <= 0.1)

m.sum_portfolio = Constraint(expr = m.NVDA + m.MSFT +m.AAPL + m.GOOGL + m.TSLA + m.AVGO + m.RBLX + m.V + m.CRM + m.DIS == 1)

m.return_floor = Constraint(expr = m.objective >= 0.01)

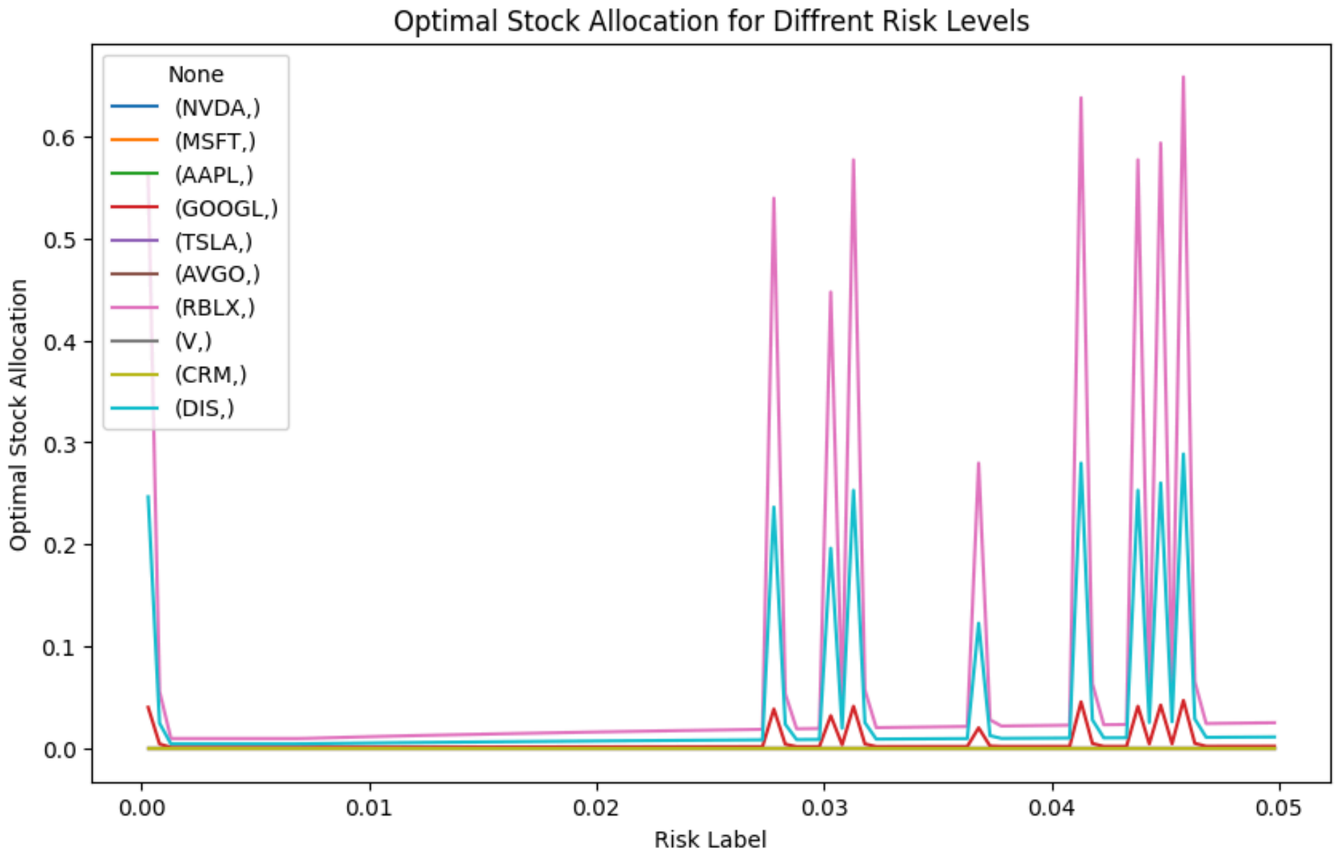
Baseline Model



Risk = 100%

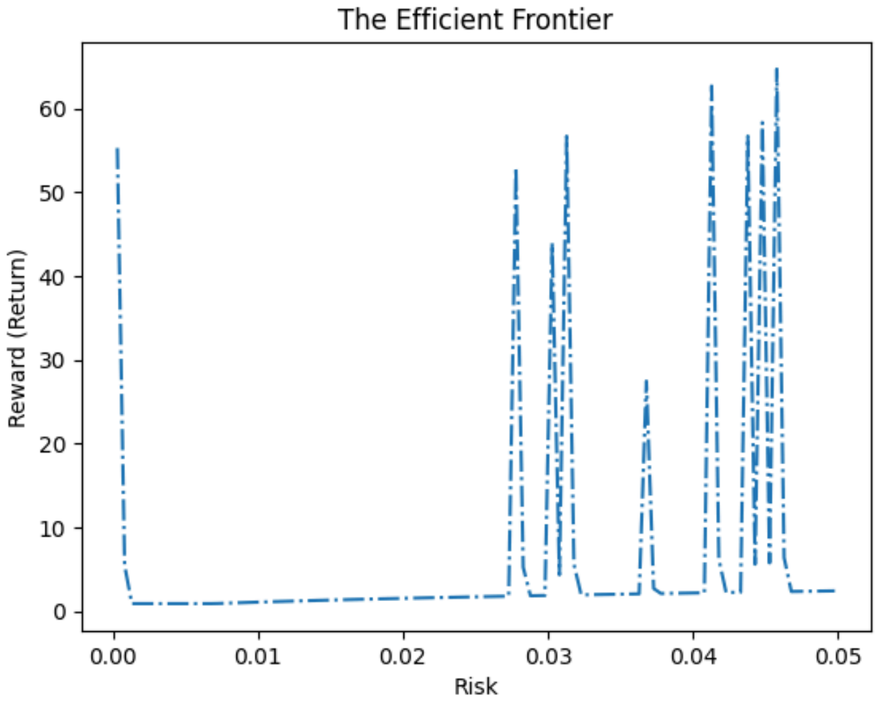
Return = 60%

Optimal Model



Risk = 5%

Return = 60%



02

JOHN BOOZMAN

U.S. Senator from Arkansas (since 2011)



PORTFOLIO COMPANIES







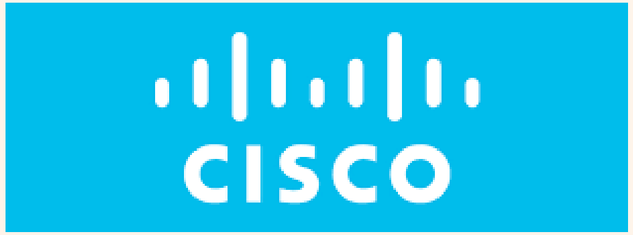














PRELIMINARY ANALYSIS

Shape of Data Set

Rows	252
Columns	90

Timeframe

January 2, 2024



December 31, 2024

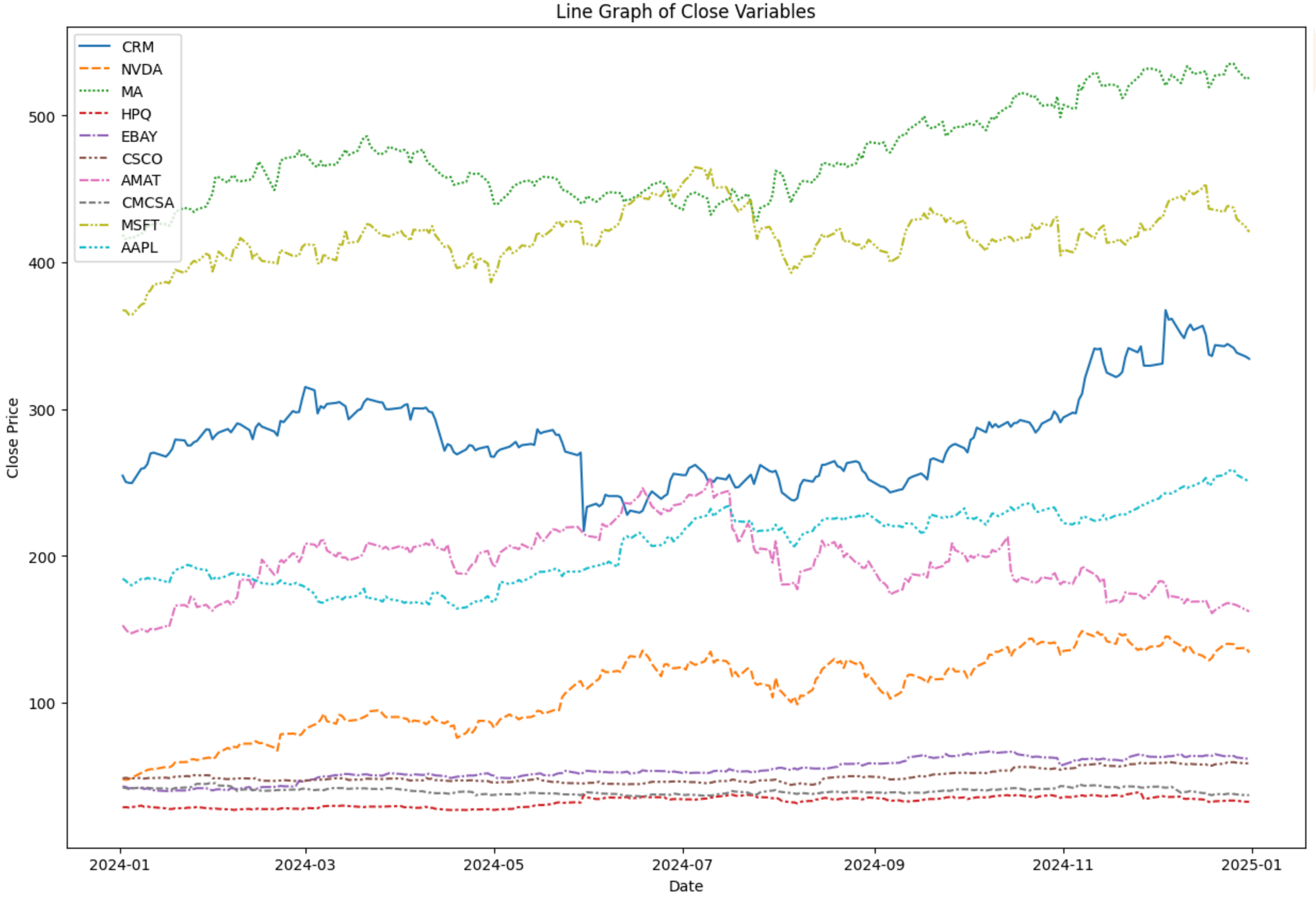
VOLATILITY & PRICE METRICS

Volatility		
Highest	CRM	$31.48
Lowest	HPQ	$3.51

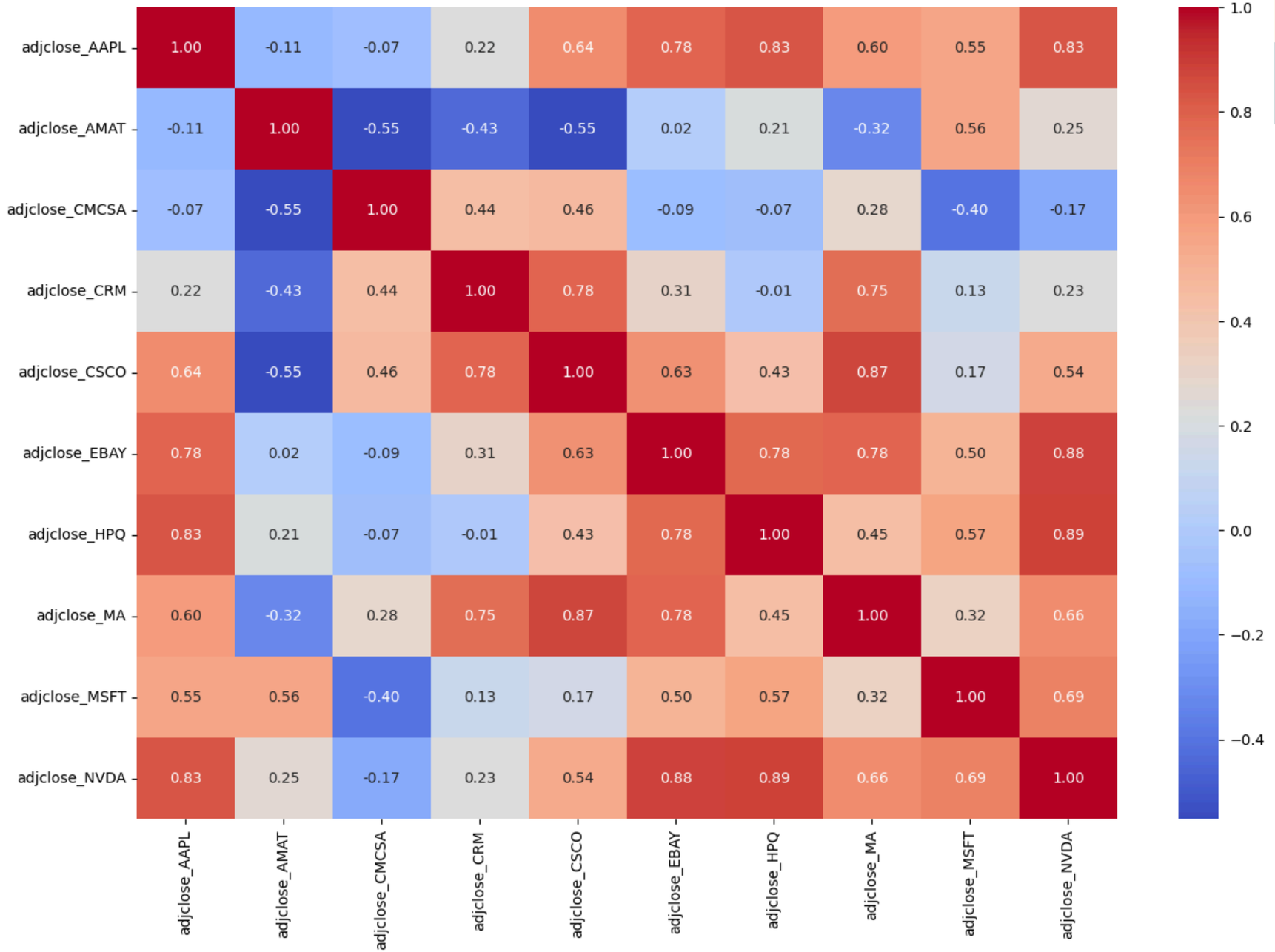


Price		
Highest	MA	$472.44
Lowest	HPQ	$32.55

DATASET GRAPH



CORRELATION MATRIX



OBJECTIVE FUNCTION

$$\text{MAX (RETURN)} = \{ 282 \times \text{CRM} + 108 \times \text{NVDA} + 472 \times \text{MA} + 32 \times \text{HPQ} + 54 \times \text{EBAY} + 50 \times \text{CSCO} + 195 \times \text{AMAT} + 40. \times \text{CMCSA} + 418 \times \text{MSFT} + 206 \times \text{AAPL} \}$$

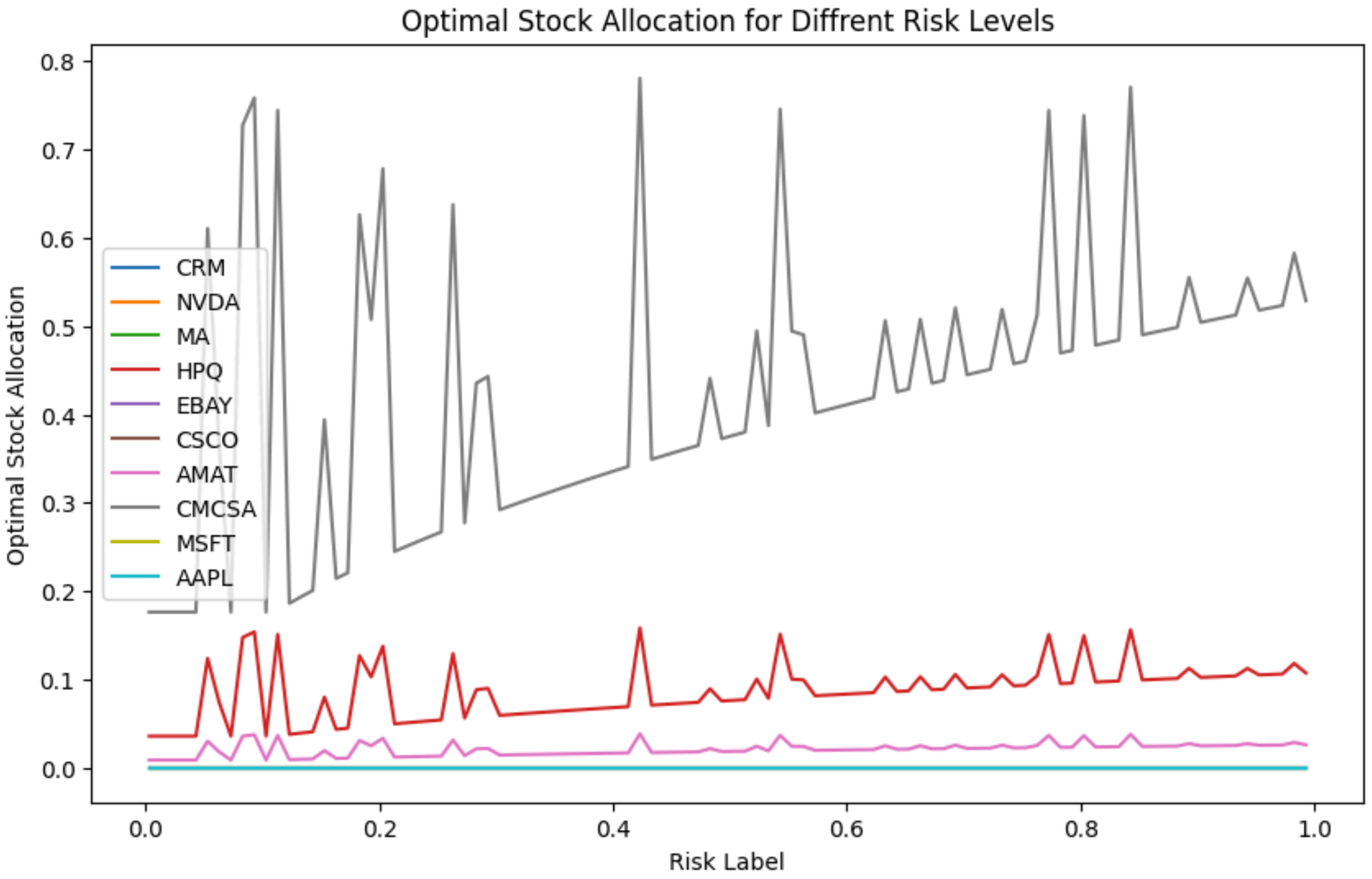
CONSTRAINTS

```
m.total_risk = Constraint(expr = m.CRM + m.NVDA + m.MA + m.HPQ + m.EBAY +
m.CSCO + m.AMAT + m.CMCSA + m.MSFT + m.AAPL == 0.1)
```

```
m.sum_portfolio = Constraint(expr = m.CRM + m.NVDA + m.MA + m.HPQ +
m.EBAY + m.CSCO + m.AMAT + m.CMCSA + m.MSFT + m.AAPL == 1)
```

```
m.return_floor = Constraint(expr = m.objective >= 0.01)
```

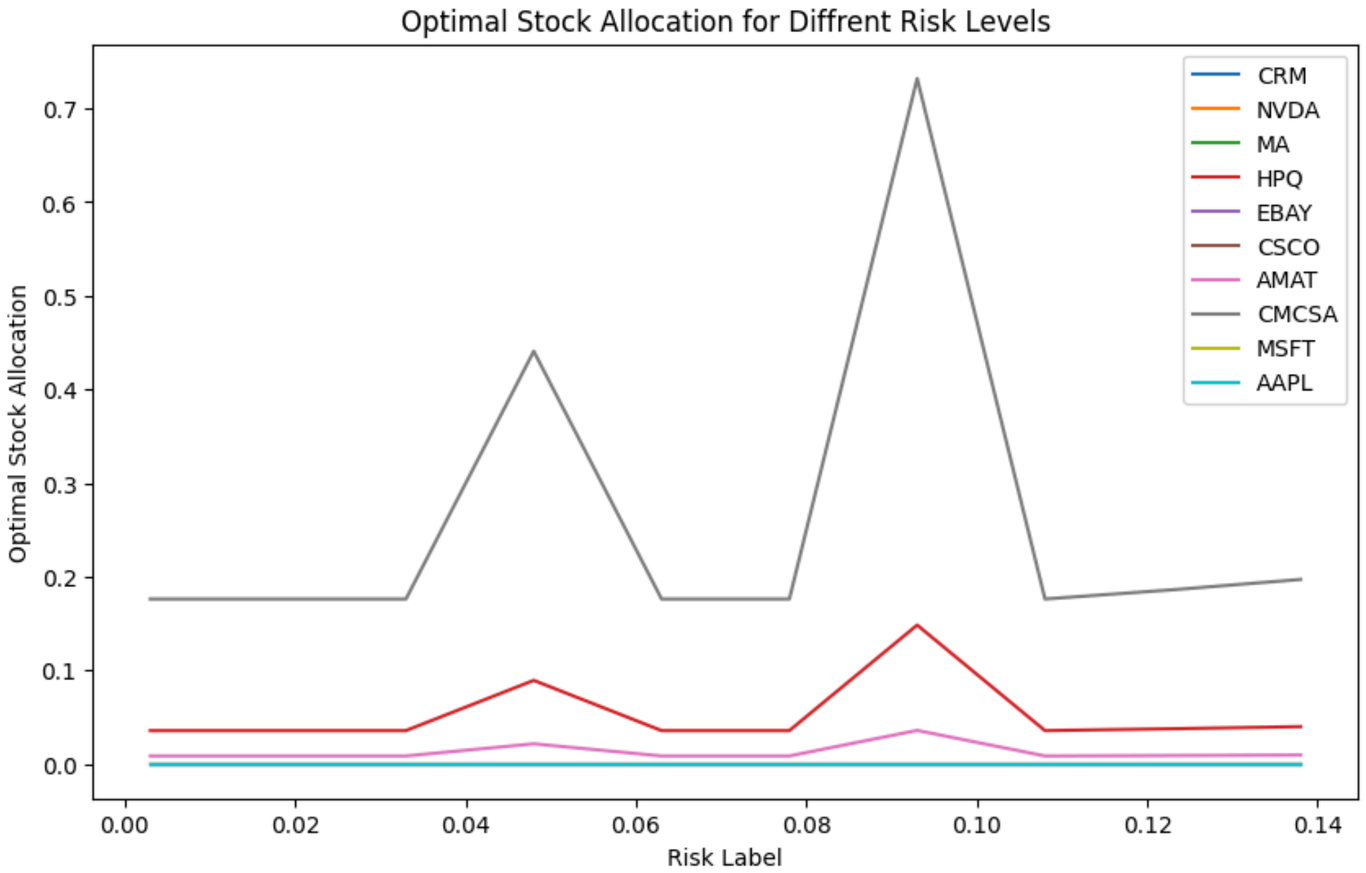
BASELINE MODEL



Risk = 100%

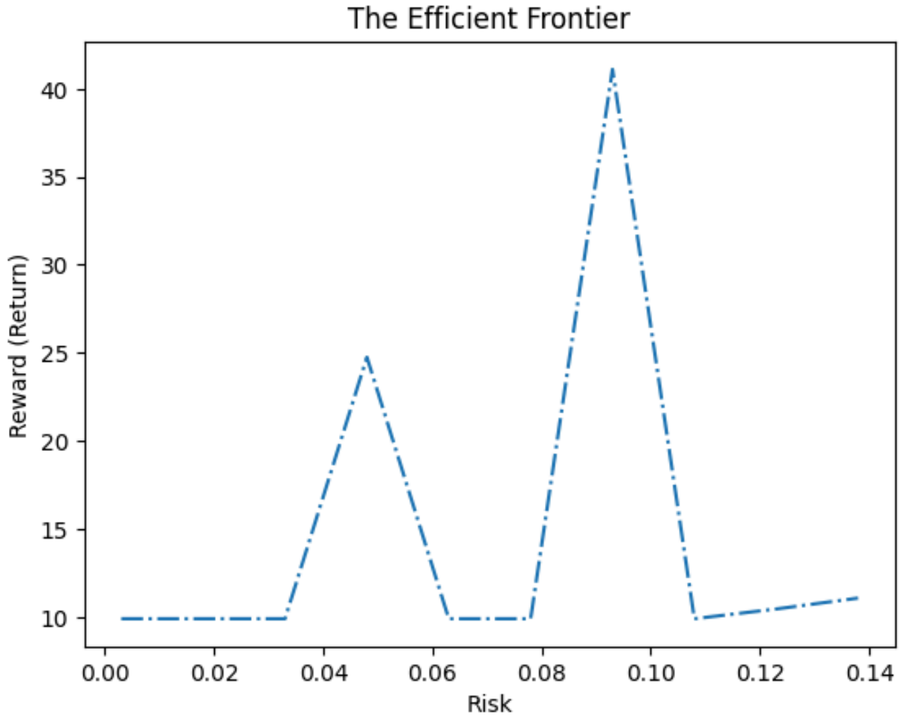
Return = 70%

OPTIMAL MODEL



Risk = 15%

Return = 40%



Final insights

Nancy Pelosi

Insights

Best performing stocks:

RBLX & DIS

John Boozman

Insights

Best performing stocks:

CMCSA & HPQ



Thank you !

Appendix

COVARIANCE MATRIX

